Text Steganography through Indian Languages Using Feature Coding Method

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***Abstract*—Steganography is the method to hide a message inside another message without drawing any suspicion to the others and that the message can only be detected by its intended recipient. With the other steganography methods such as Image, Audio, Video, a number of text steganography algorithms have been introduced. This paper presents a new approach for steganography in Indian Languages. Considering the availability of more feature code able characters and flexible grammar structure of Indian Languages this approach is able to hide a secret message in the text. The text, after replacing feature code able characters with the transformed versions, can be used as cover media. Similarly by applying a reverse method to the cover file, we show how easily the secret message can be recovered by the intended receiver.**

Keywords-Information Security, Text Steganography, Information Hiding, feature coding.

I. Introduction

The growing possibilities of modern communications need the special means of security especially on computer network. The network security is becoming more important as the number of data being exchanged on the Internet increases. Therefore, the confidentiality and data integrity are required to protect against unauthorized access. This has resulted in an explosive growth of the field of information hiding. Various methods like Cryptography, steganography, coding and so on have been used for this purpose. However, during recent years, steganography perhaps has attracted more attention than others.

Steganography is the art and science of hiding information such that its presence cannot be detected [1]. A secret message is encoded in such a manner that the existence of the information is hidden. Paired with existing communication methods, steganography can be used to carry out hidden exchanges.

The objective of steganography is to establish a secured communication in a completely undetectable manner [2] and to avoid drawing suspicion to the transmission of a hidden data [3]. It is not to keep others from knowing the hidden information, but it is to keep others from thinking that the information even exists. If a steganography method causes someone to suspect that there is secret information in a carrier medium, then the method has failed [4].

Trithemius, the author of Polygraphia and Steganographia, invented the word Steganography and the word is derived from two Greek words steganos, meaning "covered", and graphein, meaning "to write". The first written evidence about steganography being used to send messages is the Heredotous[5] story about slaves and their shaved heads.

Although steganography is an ancient subject, the modern formulation of it is often given in terms of the prisoner's problem [6]. Specifically, in the general model for steganography, illustrated in "Fig. 1", we have Alice wishing to send a secret message S to Bob. In order to do so, Alice embeds S into a cover-object C, to obtain the stego-object Ć. The stego-object Ć is then sent through the public channel. In a pure steganography framework, the technique for embedding the message is unknown to Wendy and shared as a secret between Alice and Bob. However, it is generally not considered as good practice to rely on the secrecy of the algorithm itself. In private key steganography Alice and Bob share a secret key which is used to embed the message.

In compared to other steganography such as on images [2, 4, 7], video files [8, 9], audio files [10] etc., due to the lack of large scale redundancy of information in a text file, text steganography seems to be most difficult kind of steganography [11].

This paper provides new methods for hiding information using Indian Languages based on feature coding. This method can also implemented on any language but since Indian languages are having much more characters with respect to the other language like English therefore it will be quite easier to create the words or innocuous sentences using Indian languages.

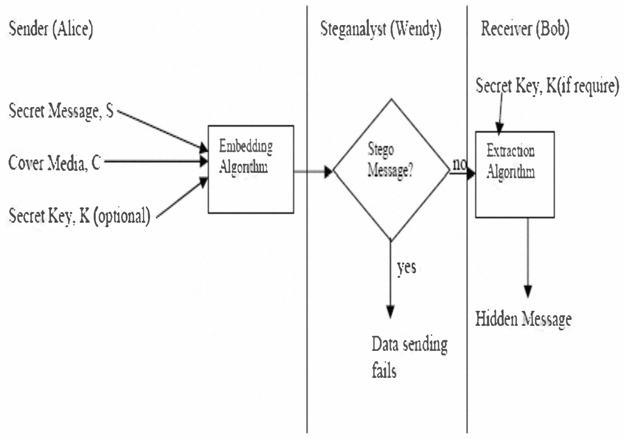

Figure 1. General Model for Steganography

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II. RELATED WORKS ON TEXT STEGANOGRAPHY

This section describes some of the related works on information hiding along with their advantages and disadvantages.

A. Syntactical Steganography

Considering the syntactic structures of a text, the syntactical steganography approach build syntactically correct sentences by using the Context Free Grammars (CFG). CFG based Mimicry [12], NICETEXT [13] comes under this category. Though, NICETEXT produces syntactically correct sentences, the output text is almost always set of ungrammatical and semantically anomalous sentences. Using this disadvantage of NICETEXT steganalysis algorithms [14] have been developed to detect the presence of hidden information in the cover file generated by NICETEXT.

B. Lexical Steganography

In lexical Steganography lexical units of natural language text such as words are used to hide secret bits. . In this approach a word could be replaced by its synonym and the choice of word to be chosen from the list of synonyms would depend upon secret bits.

C. Ontological Technique

In this method, to embed information, instead of implicitly leaving semantics intact by replacing only synonymous words an explicit model for the meaning is used to evaluate equivalence between texts. This method is also having the same disadvantage like NICETEXT that sometimes it may produce semantically incorrect texts.

D. Other Steganography Approaches

In Text Steganography by Hiding Information in Specific Character of Words [15] approach, specific characters from some particular words are selected to hide the information. For example, the first character of every alternative word hides the secret message.

Text Steganography by Line Shifting method [16, 17] is another useful approach where lines are shifted vertically to some degree. For example, lines are shifted vertically to degree say α or $-\alpha$. For α, the information is 1 and for $-\alpha$, the information is 0. This method is appropriate for printed text.

Information hiding in Random Character and Word Sequences method [18] generates a random sequence of characters or words to hide the information.

In Text Steganography by Word Shifting method [16, 19], information is hidden in the text by shifting words horizontally and by changing the distance between the words.

Text Steganography by Feature Coding method [20, 21] changes the feature or structure of the text to hide data. For example, elonging or shortening end portion of some characters, or by vertical displacement of points of characters like ‘i’, ‘j’ etc. In this method a large volume of data can be hidden in the text.

In Text Steganography by adding Open Spaces method [22], the information can be hidden by adding extra white spaces in the text.

Information can be hidden by Creating Spam Texts [18] in a HTML file. This approach uses the flexibility of HTML regarding case-sensitiveness. In HTML the tags
,
,
 are equally valid. So by changing case, one can hide information in a HTML documentation text. But in WML this method will not work since all the tags are in lower case letters.

Also, some algorithms for text steganography through Indian Languages such as Bengali and Hindi can be found [23, 24, 25, 26]

III. MOTIVATION AND PROPOSED SCHEME

We find a considerably new approach for stagonography in hindi texts in [25], where the Authors considered the structure of Hindi alphabet/words, to hide the secret message in the text by shifting the subpart of some Hindi characters towards left or right, by rotating the line within the main character of some Hindi characters, by shifting the rightmost vertical line of the parallel lines formed by combining the consonant with vowel, by adding more spaces between the subparts of medial vowel and by shifting the point of the medial vowel. The authors categorize this approach under feature coding methods.

In this paper we use feature coding for our proposed scheme, on Indian Languages. This paper proposes a new scheme on how to efficiently apply feature coding methods for embedding secret messages.

The classic way of representing a binary (secret) string of zero's and one's of length n is, as a sequence of outcome of n coin tosses; and hence use this uncertainty as the basic building block to represent the string by the sequence of outcome of any such events that has outcomes similar to a coin toss. An example using feature coded symbols from any language is given for a short illustration.

In this example, the existence of the original symbol in the text can be modeled as or represented by a zero or the existence of the symbol after feature coding is applied to it, can be represented by a one. That is, the form of existence of feature code able symbols (we call transformable from now on) throughout the text is the event and the two forms, 'original' and 'feature coded' are the outcomes of the event. Thus, one can keep a transformable symbol in its original form, to represent a zero, or can apply feature coding to that symbol to represent a one. In this way, selecting transformable symbols throughout a properly chosen cover media, and applying the above method, one can represent a sequence of zeroes and ones; the places (in the cover media) of existence of transformable symbols being the places of existence of those zeroes and ones and whether it is a zero or a one is determined by the existence or non existence of the transformable symbol in its original form.

In this paper we propose a new way of representation of the sequence of zeroes and ones (the secret message), by defining a new event class.

Here we classify the transformable symbols into several categories (the method of classification being based on the different feature coding algorithms that is applicable for the given language) and define the event as 'sustaining or remaining in a specific category during transformation of transformable symbols (from their original form to the feature coded form).

This event again has two outcomes –

"Transformation of transformable symbols, of the same category",

OR

"Transformation of transformable symbols of a different category".

The event is not independent as the 'coin toss' event in the sense that the n th invocation of the event depends on the outcome of the n-1 th invocation of the event. We show that it is possible to model an event like this in the scope of our discussion, and the dependence of the next invocation over the outcome of the previous invocation only tightly couples the invocations with no compromise to the randomness of the method.

IV. THEORETICAL AND TECHNICAL BACKGROUND

We model the event invocations and explain the encoding/representation of a binary string with the Finite State Machine (FSM) model.

We define each state as a category of the transformable symbols and define the transition function, δ as:

$\delta(q, 0) = q$

$\delta(q, 1) = \{p \mid \text{such that } p \in Q \text{ but is not } q\}$, Q being the set of states representing the categories.

That is,

if we want to encode a zero, we select the next transformable symbol from the same category and apply a transformation.

Else if we want to encode a one, we select the next transformable symbol that does not belong to the same category, but may belong to any other categories, and apply the transformation.

The term 'same category' implies the dependence of the next invocation of the event over the outcome of the previous invocation and demands justification and explanation.

As soon as we try to model our approach with a memory less system such as FSM, we can't afford to define anything by 'remembrance of the outcome of the previous event'. With no loss of generality, we can track back from the k th invocation of the event to the very first invocation, and reach to a situation where we should know the outcome of the zeroth invocation, which does not physically comply. To solve this, we fix the very first (or zeroth!) invocation for the outcome ' transformation of transformable symbols, from the same category' and then carry on.

What we exactly do is, we fix the start state of the Finite Machine!

That is, we resolve the uncertainty around choosing a particular 'q' from Q to designate as the start state and start by:

Choosing the first transformable symbol,
Identifying its category,
Applying the transformation and
Defining this category as 'q', the start state.

Now the encoding of zeros and ones proceed according to the transition function of the FSM. Interestingly, the FSM is non deterministic and during encoding, we can use this non-determinism to make the application space more and more sparse.

A. Encoding of a binary string

Zero:

To encode a zero, we choose the next symbol (transformable) from the cover media, that belongs to the same category as the previous (or first if it is the LSB of the secret message) transformed symbol. (skipping as many symbols from different categories, as may be required).

One:

To encode a one, we choose the next symbol (transformable) from the cover media, that belongs to a different category as the previous (or first if it is the LSB of the secret message) transformed symbol. (to use the non-determinism of the FSM, we can skip an arbitrary number of transformable symbols (even if they may be from different categories) and choose any symbol from among any other category).

Thus an encoding of 0110001101 can be modeled as a sequence of state changes:

$q3 \rightarrow q3 \rightarrow q6 \rightarrow q1 \rightarrow q1 \rightarrow q1 \rightarrow q1 \rightarrow q2 \rightarrow q4 \rightarrow q4 \rightarrow q3$

we can think of hiding the string in between state transitions, which can be observed as:

$q3 \rightarrow q3 \rightarrow q6 \rightarrow q1 \rightarrow q1 \rightarrow q1 \rightarrow q1 \rightarrow q2 \rightarrow q4 \rightarrow q4 \rightarrow q3$

0 1 1 0 0 0 1 1 0 1

and in this sense, we call this encoding scheme as a new text steganographic approach where the binary string (the secret message) is thus hidden between state transitions, where q_n represents operations on symbols (transformable) from category n.

Here, the first transformable symbol from the cover media was from category 3 and so, q3 is the start state, and to encode a 0, we remain at q3.

Another example of encoding a binary string that starts with a one is also given:

An encoding of 1010001101 can be modeled as a sequence of state changes:

$q5 \rightarrow q3 \rightarrow q3 \rightarrow q1 \rightarrow q1 \rightarrow q1 \rightarrow q1 \rightarrow q2 \rightarrow q4 \rightarrow q4 \rightarrow q3$

we can think of hiding the string in between state transitions, which can be observed as:

$q5 \rightarrow q3 \rightarrow q3 \rightarrow q1 \rightarrow q1 \rightarrow q1 \rightarrow q1 \rightarrow q2 \rightarrow q4 \rightarrow q4 \rightarrow q3$

1 0 1 0 0 0 1 1 0 1

Here, the first transformable symbol from the cover media was from category 5 and so, q5 is the start state, and to encode a 1, we change state to q3.

B. Decoding

step1. The intended receiver (only receiver from now on) searches for the first transformed symbol from the cover media and notes the category to which the original symbol belongs.

Step2. The receiver finds the next transformed symbol and its category.

Step3. If the categories form step 1 and step 2 matches, then the (first) decoded symbol is a zero else the (first) decoded symbol is a one.

Step4. Steps1 through 3 are repeated until the last transformed symbol in the cover media is searched for.

Thus the decoding phase is a simple symmetric reverse of the encoding phase where outcome of the different invocations of the event (during encoding) are reconstructed and the result is the binary string (the secret message) of zeroes and ones. Also, the actions of the receiver are fully deterministic in-spite of (any or many) possible use of non-determinism by the encoder during the encoding phase.

V. ADVANTAGES AND DISADVANTAGES

Allowing use of non-determinism during encoding phase immediately allows several different encoding s of the same binary string. For example another encoding of '1010001101' can be,

modeled as a sequence of state changes:

$q5 \rightarrow q6 \rightarrow q6 \rightarrow q2 \rightarrow q2 \rightarrow q2 \rightarrow q2 \rightarrow q1 \rightarrow q3 \rightarrow q3 \rightarrow q4$

we can think of hiding the string in between state transitions, which can be observed as:

$q5 \rightarrow q6 \rightarrow q6 \rightarrow q2 \rightarrow q2 \rightarrow q2 \rightarrow q2 \rightarrow q1 \rightarrow q3 \rightarrow q3 \rightarrow q4$

1 0 1 0 0 0 1 1 0 1

Since states indicate categories of transformable symbols, different encoding of the same binary string (as a sequence of state transitions) actually imply the many choices between the transformable symbols the encoder can use, throughout the cover media; thus making the transformation more and more sparse; thus reducing the density of secret message over the cover media (although not a requirement for steganographic approaches).

This method induces a very little protocol agreement overhead between the encoder (sender) and decoder (receiver).

The receiver only needs to know the categories of the transformable symbols.

VI.EXPERIMENTAL RESULTS

For hiding the message we have considered a small portion from a topic of a daily Bengali newspaper.We use a scanner program on the Bengali text and build a table of feature code able characters and their positions in the text. We group the characters according to the class of feature coding method that can be applied on them, for example, Steganography by shifting parallel lines, Steganography by point shifting, Steganography by 'ref' (a medial vowel placed at the top of the main character) and name these classes as C_F1, C_F2, C_F3 respectively, each signifying the collection of feature code able characters.

We build the FSM representing these categories and encode the binary string 10000101011111001101 1000010, according to the transition function, each transition encoding a bit. This, when decoded by the intended receiver, is the word 'BOMB'. We explain this proof of the concept experimental demonstration by, "Fig.2", "Fig.3", "Fig.4" and "Fig.5".

VII.CONCLUSION AND FUTURE RESEARCH SCOPE

In this paper we have introduced a new approach of using feature coding for Text Steganography through Indian Language. This system is taking the advantage of the existence of many different types of feature code able characters in Indian Languages. The text, after replacing feature code able characters with the transformed versions, is used as cover media. Similarly by applying a reverse method to the cover file, we have shown how easily the secret message can be recovered by the intended receiver.

Future research can be made to develop modified algorithm to decrease number of iteration which will lead to decrease the size of the cover file and the time complexity of the algorithm.

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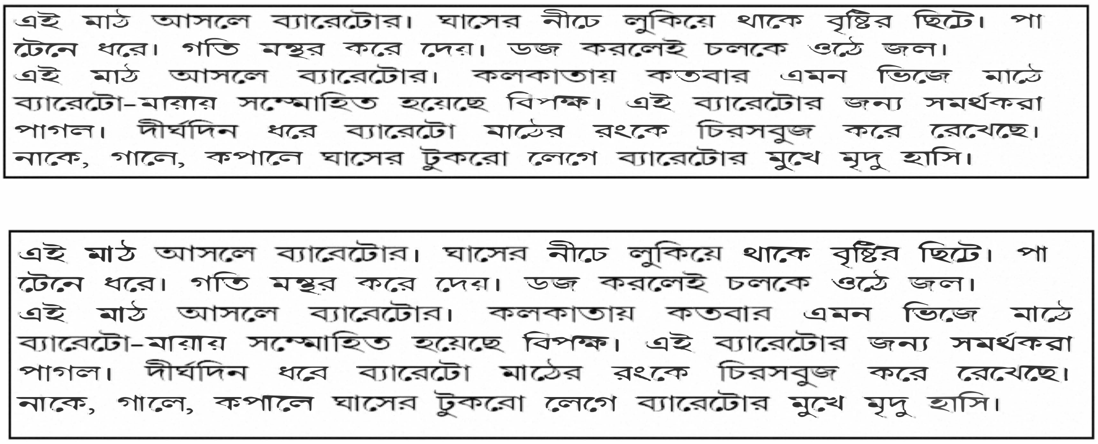

Figure 3. Cover media generated by applying our algorithm to the original text file.

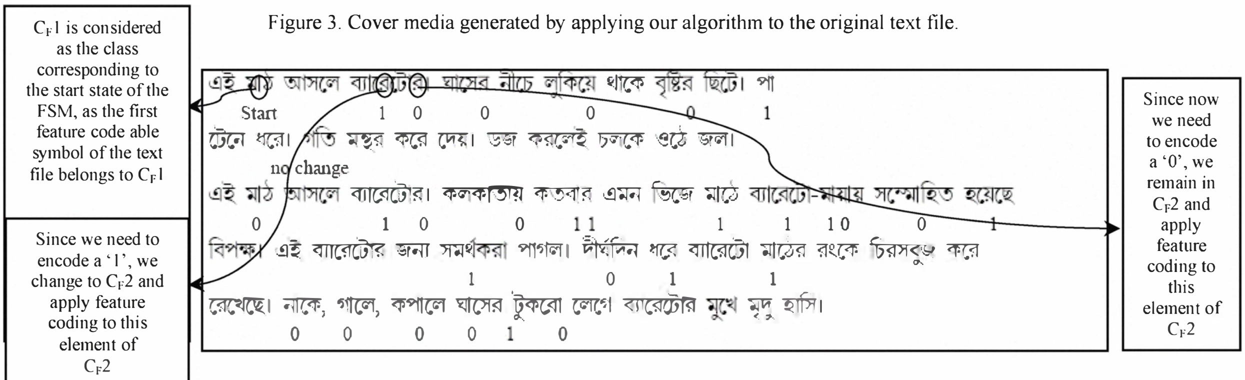

Figure 4. Illustration of working methodology of the algorithm.

$C_F1 \rightarrow C_F2 \rightarrow C_F2 \rightarrow C_F2 \rightarrow C_F2 \rightarrow C_F1 \rightarrow C_F1 \rightarrow C_F2 \rightarrow C_F2 \rightarrow C_F2 \rightarrow C_F1 \rightarrow C_F2 \rightarrow C_F1 \rightarrow C_F2 \rightarrow C_F1 \rightarrow C_F2 \rightarrow C_F1 \rightarrow C_F2 \rightarrow C_F3 \rightarrow C_F3 \rightarrow C_F2 \rightarrow C_F1 \rightarrow C_F1 \rightarrow C_F1 \rightarrow C_F1 \rightarrow C_F2 \rightarrow C_F2$

Figure 5. moves of the FSM for encoding the word 'BOMB'